

KUSH AGARWAL

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[linkedin](#) , [github](#) & [Leetcode](#)

Education:

Degree: B.Tech in Computer science, specialization in AI and ML

University: GITAM, Visakhapatnam

Timeline: 2025-2029

Open source contribution:

- **Kornia:** successfully merged a PR with Kornia organization and navigated their strict CI/CD and review workflows. I wrote documentation for ImageSequential and AugmentationSequential. ([PR](#))
- **Gammapy:** Recognized as an [official contributor](#) to the V2.1 release. Optimized core N-dimensional mapping algorithms by implementing dynamic tensor slicing with NumPy, delivering a ~500x performance speedup for irregular geometry cropping. ([PR](#) and [citation](#))

Technical projects:

- **Word embedding model** : I have built a word embedding model using PyTorch from scratch without using any APIs. I trained it using the WIKItorch-103 dataset. Built for experimental learning purposes. ([Link](#))
- **Analysis-XYZ** :Developed a full-stack data analysis web tool using Python and JavaScript. Focused on end-to-end architecture, from routing to real-world deployment and scaling for live users. I have successfully deployed it and its [live](#) . The main purpose of building it was to learn how to scale and deploy for real world users.
- **Agentic chat support system** : LangChain-powered agentic support system with real tool-calling, multi-turn memory, and dynamic intent routing deployed live on HuggingFace Spaces. [Live](#)

Technical skills:

- **Languages** : Python, C

- **Libraries/Framework:** NumPy, Pandas, Flask, Spacy(NLP), Langchain, LangGraph
- **Tech and workflows:** GIT, GitHub